

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

1

1. Physical Definition: Reconfiguration of Internal Parameters via Structural Isomerization

We define that when the material absorbs magnetic energy E_{mag} and the **Structural Isomerization Rate (α)** progresses, the material's **Complex Permeability (μ^*)** or **Complex Impedance (Z^*)**—derived from the internal potential—undergoes a non-linear transformation.

The critical distinction here is that the **Isomerization Rate (α)** is a direct function of the "**Cumulative Magnetic Exposure.**"

2. Core Formulation: Conversion of Magnetic Energy into Potential States

In conventional electromagnetics, magnetic flux change results in **Induced Electromotive Force (V_{ind})**. In this material, however, that energy is deterministically converted into **Isomerization Energy (U_{iso})**.

The resultant shift in the readout parameters (e.g., **Resistance R_e** or **Reactance X**), denoted as ΔZ , is characterized as a **Discrete Step Response** as follows:

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

2

(Mathcad-Compatible Formulation)

A. Input & Threshold Constants

$$k_B := 1.3806 \cdot 10^{-23} \quad [\text{J/K}]$$

$$T_{char} := 28.15 \quad [\text{K}]$$

$$E_{act} := k_B \cdot T_{char} \quad [\text{J}] \quad (\text{Activation Barrier})$$

B. Energy Integration

$$E_{mag} := \int V_{ind}(t) \cdot I(t) dt \quad [\text{J}]$$

C. Quantized Impedance Shift (The Discrete Solution)

$$\Delta Z(E_{mag}) := \sum_{n=1}^N \delta_{step} \cdot \Phi(E_{mag} - n \cdot E_{act})$$

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023

IPL 知的財産研究所村上光治むらかみこうじ
IPL Intellectual Property Institute Kouji Murakami

F A X : 0 6 7 6 3 5 2 8 1 5

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

3

Technical Commentary:

Each time the incident magnetic energy exceeds the **Activation Barrier** (E_{act})—equivalent to the process-defined **28.15 K**—the internal domains of the material undergo a "unit-cell" isomerization. This results in a **quantized, staircase-like (digital) shift** in the complex impedance matrix.

Unlike linear sensors, this material functions as a **Structural Integrator**, where the energy is "locked" into a permanent physical state, allowing for the precise dosimetry of magnetic exposure through the analysis of ΔZ .

3. Mathematical Execution: Derivation of the Isomerization Transfer Function

To validate the research empirically, the following sequence defines the transition from external magnetic flux to the internal structural "locked" state.

(Part A: Fixed Process Parameters (Material Constants))

These values represent the "Structural Memory" embedded during the manufacturing process.

- $E_{act} = k_B \cdot 28.15$ [J] (Characteristic Activation Barrier)
- $\Gamma = 1.2$ (Cooperativity Coefficient / Slope Factor)
- $Z_{baseline} = 50 + 0j$ [Ohm] (Pre-exposure State)
- $\Delta Z_{unit} = 0.5 + 0.1j$ [Ohm] (Quantized shift per isomerization event)

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023

IPL 知的財産研究所村上光治むらかみこうじ
IPL Intellectual Property Institute Kouji Murakami

F A X : 0 6 7 6 3 5 2 8 1 5

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

4

(Part A: Fixed Process Parameters (Material Constants))

These values represent the "Structural Memory" embedded during the manufacturing process.

- $E_{act} := k_B \cdot 28.15$ [J] (Characteristic Activation Barrier)
- $\Gamma := 1.2$ (Cooperativity Coefficient / Slope Factor)
- $Z_{baseline} := 50 + 0j$ [Ohm] (Pre-exposure State)
- $\Delta Z_{unit} := 0.5 + 0.1j$ [Ohm] (Quantized shift per isomerization event)

(Part B: Incident Energy Quantification (Dynamic Input))

The magnetic potential that initiates the isomerization.

- $\Phi_{dot}(t) := A_{eff} \cdot \frac{dB}{dt}$ [Wb/s] (Magnetic flux change rate)
- $E_{incident} := \int |\Phi_{dot}(t) \cdot I(t)| dt$ [J] (Integrated incident energy)

[Part C: The Transfer Function (The Solution for α)]

The core equation that solves for the isomerization progress rate. This is the bridge between magnetic input and physical output.

$$\alpha(E_{incident}) := \frac{1}{1 + \exp\left(-\Gamma \cdot \frac{E_{incident} - E_{act}}{E_{act}}\right)}$$

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023
IPL 知的財産研究所 村上光治 むらかみ こうじ
IPL Intellectual Property Institute Kouji Murakami
FAX : 0 6 7 6 3 5 2 8 1 5

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

5

(Part D: Observable Solutions (Final Output Parameters))

The final equations used to calculate the measurable scientific evidence.

- $Z_{result} := Z_{baseline} + \Delta Z_{unit} \cdot \text{round}\left(\frac{E_{incident}}{E_{act}}, 0\right)$

Solution: The staircase-like impedance shift representing the cumulative dose.

- $V_{suppressed} := V_{theory} \cdot (1 - \alpha(E_{incident}))$

Solution: The resultant voltage after the back-EMF is absorbed into the structural lattice.

- $U_{stored} := E_{incident} \cdot \alpha(E_{incident})$

Solution: Total magnetic energy successfully converted into structural potential energy.

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023
IPL 知的財産研究所村上光治むらかみこうじ
IPL Intellectual Property Institute Kouji Murakami
FAX : 0676352815

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

6

4. Resultant Physical Properties: Discrete Structural Fingerprinting

This is the final solution where the magnetic energy is permanently "locked" into the material. The results are defined not as transient losses, but as **Discrete State Changes**.

[The Solution for Quenched Potential]

The Back-EMF is not merely "suppressed"; it is functionally eliminated as the magnetic flux is consumed by the isomerization energy barrier.

$$V_{residual} = V_{theory} \cdot [1 - \alpha(E_{mag})]$$

(When $\alpha \rightarrow 1$, $V_{residual} \rightarrow 0$: The energy is entirely internalized.)

[The Solution for Structural Memory (The Readout)]

The post-exposure state is a deterministic shift in the Complex Impedance Matrix. This is the "**Unexpected Effect**" that proves the product-by-process uniqueness.

$$Z_{final} = Z_{base} + \sum_{n=1}^N \Delta Z_n \cdot \Theta(E_{mag} - n \cdot E_{act})$$

- Θ : Heaviside step function (Ensuring discrete, staircase jumps).
- Z_{final} : The permanent, non-volatile structural record of the total exposure.

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023
IPL 知的財産研究所村上光治むらかみこうじ
IPL Intellectual Property Institute Kouji Murakami
F A X : 0 6 7 6 3 5 2 8 1 5

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

7

Supplementary Note: Detailed Mechanism of Discrete Structural Fingerprinting

1. Non-Radiative Energy Internalization

In standard Maxwellian physics, a change in magnetic flux ($\Delta\Phi$) must result in an induced electromotive force (Back EMF). However, in this process, the energy is not "radiated" or "dissipated" as an electrical signal.

- **The Mechanism:** The magnetic energy is consumed as **Work** to overcome the specific activation barrier ($E_{act} \approx 28.15 \text{ K}$).
- **The Result:** The incident flux is effectively "quenched" and internalized into the lattice potential, leading to a state where $V_{back_emf} \approx 0$ while $\alpha \rightarrow 1$.

2. Quantized Phase Transition (The Staircase Response)

The shift in physical properties is not a linear slope but a series of **discrete jumps**. This is the core of the "Unexpected Effect."

- **The Threshold Logic:** Because the manufacturing process embeds a uniform energy barrier, the material remains inert until the incident energy reaches $n \cdot E_{act}$.
- **The Digital Nature:** Upon reaching the threshold, a localized domain undergoes a collective spin-lattice reconfiguration (Isomerization). This ensures that the record of exposure is **quantized**, allowing for deterministic counting of magnetic stress units rather than ambiguous analog estimation.

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023
IPL 知的財産研究所村上光治むらかみこうじ
IPL Intellectual Property Institute Kouji Murakami
FAX : 0676352815

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

8

3. Permanent Structural Memory (The Fingerprint)

Unlike Hall-effect or induction sensors which require constant power and provide only real-time data, this material acts as a **Non-Volatile Structural Memory**.

- **The Post-Exposure State**: Once the isomerization occurs, the material is "locked" into a higher potential state.
- **The Measurable Evidence**: This state change is permanently reflected in the **Complex Impedance Matrix (Z^*)**. The shift from Z_{base} to Z_{final} is irreversible under standard environmental conditions, providing a physical "fingerprint" of the maximum magnetic exposure.

4. Scientific Superiority in Dosimetry

This mechanism provides the first rigorous solution for **Bio-magnetic Dosimetry**.

- **Inherent Precision**: By measuring the total impedance shift ΔZ , we can back-calculate the cumulative energy exposure with semiconductor-level precision.
- **Process-Product Link**: The accuracy of the measurement is directly guaranteed by the **Manufacturing Process (Product-by-Process)**, as the E_{act} threshold is a physical constant inherent to the material's synthesized structure.

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023
IPL 知的財産研究所村上光治むらかみこうじ
IPL Intellectual Property Institute Kouji Murakami
F A X : 0 6 7 6 3 5 2 8 1 5

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

9

Comparative Analysis: Discontinuity from Conventional Sensors

1. Reversibility vs. Irreversibility

- **Existing Technology (Hall/MR Sensors):** These are "Transient Sensors." They provide a proportional real-time output that disappears when the field is removed.
- **This Research (FII Material):** This is a "Structural Memory Material." The isomerization remains "locked" even after the trigger field is gone. The response is **permanently embedded** as a structural fingerprint.

2. Linear Sensitivity vs. Quantized Step-Response

- **Existing Technology:** Operates on a continuous linear scale, vulnerable to noise and drift at low-level or high-gradient field inputs.
- **This Research:** Operates on a **Discrete Threshold Mode**. By setting the $E_{act} \approx 28.15$ K, it ignores sub-threshold noise and only records significant magnetic impacts that exceed the "Knee" characteristic. This ensures **zero-drift accumulation**.

3. Power Dependency vs. Self-Actuating Dosimetry

- **Existing Technology:** Requires an external power source to bias the sensor and a CPU to integrate the data over time.
- **This Research:** The material itself performs the **Physical Integration**. The isomerization is self-actuating; the energy of the magnetic field provides the work for the phase transition. It is an **autonomous, power-free dosimeter**.

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023
IPL 知的財産研究所村上光治むらかみこうじ
IPL Intellectual Property Institute Kouji Murakami
FAX : 0 6 7 6 3 5 2 8 1 5

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

1 0

Technical Summary: The Functional Discontinuity and Final Solution

1. The Objective

To realize a Magnetic Energy Absorption Chip (MEAC) based on "Field-Induced Isomerization (FII)." Unlike inductive sensors, this material internalizes high-gradient magnetic energy without radiative emission.

2. The Critical Limitation of Existing Technology

Conventional dosimetry, such as Geiger counters or Photon counters, is fundamentally incapable of quantifying the sub-dermal biological impact caused by intense magnetic fields (e.g., Superconducting Magnets). These devices are "blind" to non-ionizing magnetic stress and cumulative bio-magnetic damage.

3. The Definitive Solution: Magnetic Energy Neutralization

This technological gap is uniquely resolved through the implementation of this FII-based Sensor Chip. By functioning as a physical "absorber," the chip does not merely observe the field—it internalizes the magnetic stress that would otherwise impact the human body. This allows for the mitigation and precise quantification of bio-magnetic harm, providing a deterministic solution that is absent in current Quantum Mechanics and Electromagnetics.

4. Scientific Breakthrough

This chip achieves what state-of-the-art physical sciences have yet to address: the conversion of hazardous, transient magnetic flux into a stable structural record. It stands as the only viable solution for the safety management of personnel in high-gradient magnetic environments.

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023
IPL 知的財産研究所村上光治むらかみこうじ
IPL Intellectual Property Institute Kouji Murakami
FAX : 0676352815

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

1 1

Medical Necessity: The Rationale for Bio-magnetic Dosimetry

1. Limitation of Current Safety Standards

Current medical safety guidelines (such as ICNIRP) primarily focus on "Thermal Effects" (SAR) or "Induced Nerve Stimulation." However, they lack a methodology to quantify the cumulative structural stress at the cellular or molecular level caused by chronic or high-gradient magnetic exposure.

The Problem: There is no "Biological Dose" unit for magnetism equivalent to the "Sievert (Sv)" for radiation.

2. Sub-dermal Magnetic Impact (Micro-Mechanical Stress)

High-gradient fields from superconducting magnets exert physical force on paramagnetic and diamagnetic molecules within the body (e.g., Hemoglobin, Enzyme cofactors, Lipid bilayers).

Mechanism of Harm: This is not "radiation damage" but "Structural Displacement Stress." Chronic exposure can disrupt ion channel gating and enzymatic reaction rates.

The Need for FII: Conventional sensors measure the field around the body, but the FII Sensor Chip—designed to mimic sub-dermal absorption—quantifies the actual energy internalized by the tissue.

3. Preventive Medicine and Long-term Risk Management

For medical staff and researchers working near MRI or superconducting systems, the long-term cumulative impact remains an "unknown risk."

Clinical Value: By providing a "Cumulative Magnetic Dose Record," this chip enables longitudinal health monitoring. It transitions magnetic safety from "simple avoidance" to "Evidence-Based Exposure Management."

4. Bridging Physics and Bio-electromagnetics

The conversion of magnetic flux into a structural fingerprint (as defined in our physical model) provides the first objective data that medical professionals can use to correlate specific magnetic "doses" with biological symptoms or cellular changes.

〒536-0023 大阪市城東区東中浜 8-7-9-606

8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023

IPL 知的財産研究所村上光治むらかみこうじ

IPL Intellectual Property Institute Kouji Murakami

F A X : 0 6 7 6 3 5 2 8 1 5

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

1 2

Medical Necessity based on Structural Energy Absorption

1. The Blind Spot: Cumulative Kinetic Stress

Modern medicine lacks a method to quantify the "Cumulative Kinetic Stress" exerted on biomolecules by high-gradient magnetic fields. Unlike ionizing radiation, which causes DNA strand breaks, strong magnetism induces mechanical torque and structural distortion in paramagnetic proteins (e.g., Hemoglobin, Ferritin).

The Specific Need: Conventional sensors only measure real-time field strength (B), ignoring the accumulated energy that causes molecular-level displacement over time.

2. Sub-dermal Energy Deposition (Internalization)

Biological tissues do not merely "exist" in a magnetic field; they interact with it. The high-gradient fields from superconducting magnets create Eddy currents and Lorentz forces within sub-dermal layers.

The Role of FII: Since the FII sensor chip is designed to absorb and internalize this energy (converting flux into structural potential), it serves as a "Biological Phantom." It quantifies the exact amount of magnetic energy that is "deposited" into human tissue—a value that remains invisible to Geiger counters or static Hall sensors.

3. Defining the "Magnetic Dose" for Personnel Safety

For researchers and medical staff, the "harm" is the cumulative result of these localized structural displacements.

The Solution: By using the FII chip's staircase-like impedance shift as a proxy, medical professionals can finally establish a "Magnetic Exposure Dose" (equivalent to a magnetic Sievert). This allows for the transition from subjective symptom reporting to objective, deterministic safety management.

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023
IPL 知的財産研究所村上光治むらかみこうじ
IPL Intellectual Property Institute Kouji Murakami
FAX : 0676352815

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

1 3

Medical Necessity: The "Biological Phantom" Approach

1. Functional Equivalence to Human Tissue

Human molecules (such as Hemoglobin and ion channels) are not static; they experience structural displacement under high-gradient magnetic fields. Our sensor chip mimics this biological interaction by internalizing magnetic flux through "Field-Induced Isomerization (FII)."

2. Quantifying the "Absorbed Dose"

Conventional sensors (Hall/Geiger) only measure external field levels, providing no data on how much energy was actually deposited into the body. This FII chip serves as a Biological Phantom, acting as a surrogate that absorbs and "records" the cumulative kinetic stress that would otherwise impact sub-dermal tissues.

3. Deterministic Safety through Energy Substitution

Because the sensor consumes the energy in a manner identical to the internal potential changes in human cells, the resulting quantized impedance shift provides the only objective evidence of the "Physical Dose" received by personnel. This transforms magnetic safety from vague estimation to deterministic dosimetry.

〒536-0023 大阪市城東区東中浜 8-7-9-606
8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023
IPL 知的財産研究所 村上光治 むらかみ こうじ
IPL Intellectual Property Institute Kouji Murakami
FAX : 0676352815

Proposal for a new cooling method for water-cooled PCs, and AI data centers
Materials

Confirmed date obtained

2026.04.14 Tuesday Around am11:45 First acquisition

Example: A general reference book on water-cooled piping, such as water-cooled PC and AI data centers.

Magnetic Energy Absorption Chip (MEAC)

1 4

Example

Power, not GPUs, is the limiting factor for AI data centers

<https://www.rolandberger.com/en/Insights/Publications/Power-not-GPUs-is-the-limiting-factor-for-AI-data-centers.html>

From Power Distribution to Heat Concentration at the Rack Level

"The transition to 800-volt direct current (VDC) power distribution represents more than an incremental efficiency improvement. It fundamentally changes where heat is generated and concentrated within a data center. Traditional AC architectures distribute power through bulky copper cabling, requiring multiple conversion stages from the utility feed to the chip. Each conversion creates heat, scattering thermal loads across power distribution units, uninterruptible power supplies, and rack-mounted power shelves throughout the facility."

Medical Necessity: The "Biological Phantom" Approach

1. Functional Equivalence to Human Tissue

Human molecules (such as Hemoglobin and ion channels) are not static; they experience structural displacement under high-gradient magnetic fields. Our sensor chip mimics this biological interaction by internalizing magnetic flux through "Field-Induced Isomerization (FII)."

2. Quantifying the "Absorbed Dose"

Conventional sensors (Hall/Geiger) only measure external field levels, providing no data on how much energy was actually deposited into the body. This FII chip serves as a Biological Phantom, acting as a surrogate that absorbs and "records" the cumulative kinetic stress that would otherwise impact sub-dermal tissues.

3. Deterministic Safety through Energy Substitution

Because the sensor consumes the energy in a manner identical to the internal potential changes in human cells, the resulting quantized impedance shift provides the only objective evidence of the "Physical Dose" received by personnel. This transforms magnetic safety from vague estimation to deterministic dosimetry.

〒536-0023 大阪市城東区東中浜 8-7-9-606

8-7-9-606 Higashinakahama, Joto-ku, Osaka 536-0023

IPL 知的財産研究所村上光治むらかみこうじ

IPL Intellectual Property Institute Kouji Murakami

F A X : 0 6 7 6 3 5 2 8 1 5